

Git and GitHub

November 16, 2025

Agenda

- What is Git?
- Installing Git
- Git basic workflow
- What is GitHub?
- Create a GitHub repo
- Push / Pull / Clone
- Branching and merging
- Collaboration workflow

What is Git?

- Distributed version control system.
- Tracks changes across the project.
- Enables team collaboration safely.

Installing Git

Windows: `choco install git`

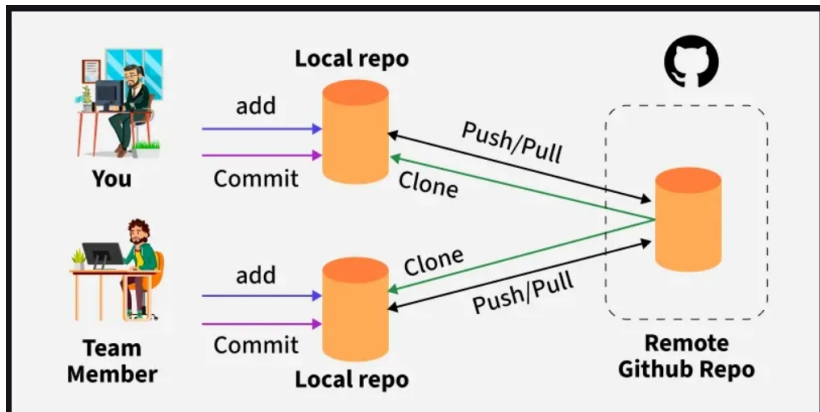
macOS: `brew install git`

Linux: `sudo apt install git`

Check installation:

```
git --version
```

Git Workflow Overview



Initialize a Repository

Commands

```
git init
```

```
git status
```

Add and Commit

Commands

```
git add .
```

```
git commit -m "Initial commit"
```

What is GitHub?

- Cloud hosting for Git projects.
- Issues, pull requests, reviews.
- Backups and collaboration.

Create a GitHub Repository

Steps:

- 1 New repository
- 2 Enter name and description
- 3 Choose visibility

Clone a Repository

Command

```
git clone URL
```

Pull and Push

Pull latest

```
git pull origin main
```

Push changes

```
git push origin main
```

Branches

Commands

```
git branch feature-x  
git switch feature-x
```

Merge

Commands

```
git switch main
```

```
git merge feature-x
```

Pull Requests

Used for:

- Code review
- Comments and improvements
- Merging contributions

Team Workflow

- ❶ Clone or fork the repo
- ❷ Create a branch
- ❸ Commit changes
- ❹ Push to GitHub
- ❺ Open a Pull Request

Summary

- Git tracks versions
- GitHub hosts repositories
- Commit → Push → Pull → Branch → Merge

Thank you!